

Good (Enough) Practices for Reproducible Research: an Introduction to Git and GitHub

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Plan

- Introduction
- Motivation
- Version Control with Git and GitHub
- More good practices
- Learning resources
- Q&A



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RSM, Erasmus University

Statistics, Transparency, and Rigor Editor
Psychological Science

Background:

- PhD in Marketing (Tilburg University)
- MSc in Econometrics and Mathematical Economics (Tilburg University)

Interests:

- (Visual) Attention
- Decision Making
- Bayesian Statistics
- Verifiable Science
- Computational Reproducibility

Good practices*
help researchers
efficiently allocate
their attention

*computing practices, open science practices

Project database › ReproHub: Enhanced Replicability and Open Science Education for Research with Secondary and Sensitive Data

ReproHub: Enhanced Replicability and Open Science Education for Research with Secondary and Sensitive Data

Despite continuous investments into open science infrastructure, research materials—like data and code—are hard to find, and even experts struggle to reproduce research results. ReproHub addresses these challenges with four deliverables to support researchers and other stakeholders in finding, documenting, and verifying research output: ReproBuddy (an AI tool to improve reproducibility packages), ReproRun (a system to validate results), OpenScienceRadar (a browser extension making open resources more visible), and OpenScienceHub.nl (a platform with accessible educational material). By building on ODISSEI and Tilburg Science Hub—two Dutch open science infrastructure components—ReproHub seeks to strengthen transparency and public trust in science.

<https://zenodo.org/records/20340149>

Characteristics

Status

In progress

Duration

2026 – 2030

Research programme

Open Science Infrastructuur

File number

500.040.2559

Grant ID

<https://doi.org/10.61686/RDRNY79776>

Discipline

Motivation

(i.e., what we're trying to achieve and what we're trying to avoid)



**USE THE
RIGHT TOOL,
FOR THE
RIGHT JOB,
IN THE
RIGHT WAY.**

(some) Research goals

- Collaborate with colleagues
- Contribute to ongoing projects
- Share your work with the world
- Easy to reproduce your own results months later

Reproducibility?

A study is reproducible if you can take the **original data** and the **original computer code** used to analyze the data and reproduce **all numerical findings** from the study.

Reproducibility or replicability?

The terms are not used consistently across different fields (<https://www.ncbi.nlm.nih.gov/books/NBK547546/>):

- A: The terms are used with no distinction between them.
- B1: “Reproducibility” refers to instances in which the original researcher's data and computer codes are used to regenerate the results, while “replicability” refers to instances in which a researcher collects new data to arrive at the same scientific findings as a previous study.
- B2: “Reproducibility” refers to independent researchers arriving at the same results using their own data and methods, while “replicability” refers to a different team arriving at the same results using the original author's artifacts.

Reproducibility Objective

Reproduce **ALL** results and **NOTHING BUT** results included in:

- Current version of the paper
- Previous versions

as fast and easy as possible.

Best case scenario

Step 1

Step 2

Step 3

Cleaning /
processing

Analysis

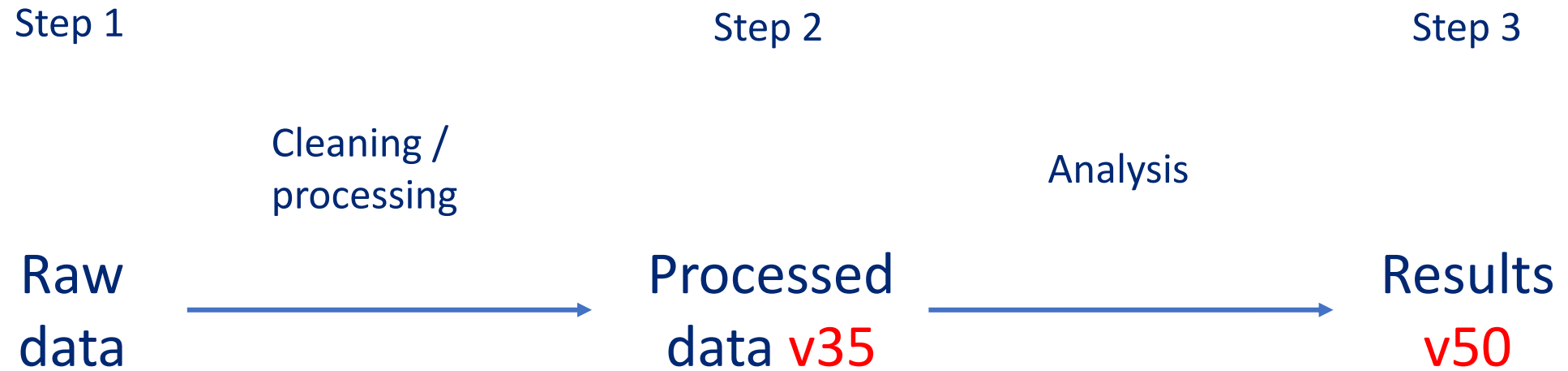
Raw
data

Processed
data

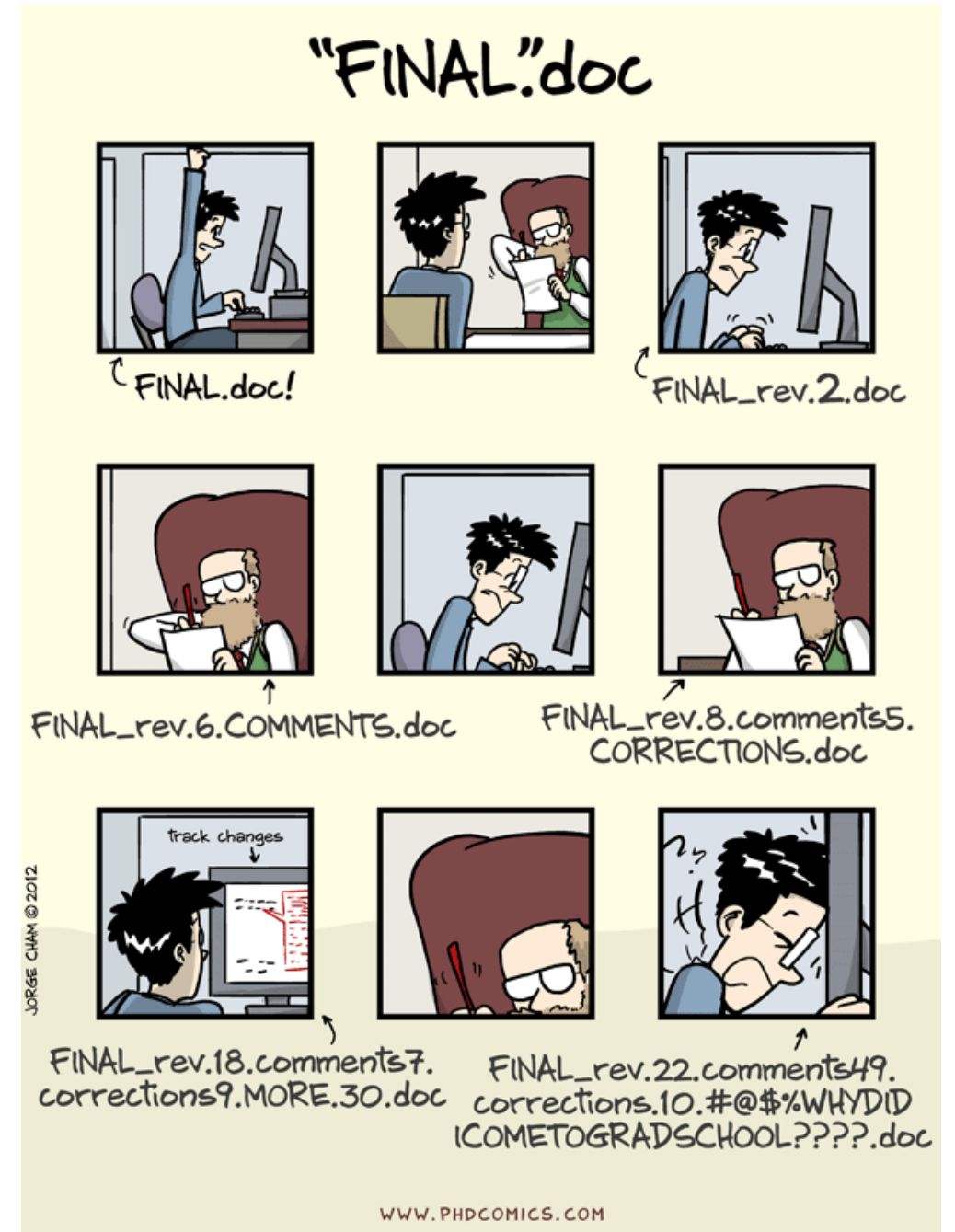
Results



Most likely scenario



What is “final”?



Things that can go wrong

- Your PC/laptop/external HD explodes
- Overwriting files, but also not overwriting files
- Forgetting which of the “final” files is really final
- Change file X, but forget to update all the other files/results that depend on it
- Other examples?



Things that can go wrong

- Your PC/laptop/external HD explodes
- Overwriting files, but also not overwriting files
- Forgetting which of the “final” files is really final
- Change file X, but forget to update all the other files/results that depend on it
- and many other examples

Version control with Git and GitHub

What is Git?

- Formal version control system
- Developed by Linus Torvalds
 - Used to manage the source code for Linux
- Tracks content:
 - Source code
 - Data analysis projects
 - Websites
 - Presentations
 - Manuscripts

What is GitHub?

- A home for git repositories
- Interface for exploring public git repositories
- A 'safe place' to keep code and data

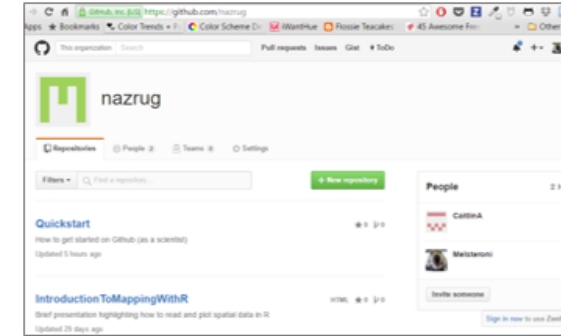
Why use GitHub?

- Facilitates
 - Exploring code
 - Tracking issues
 - Learning from others
- Lowers the barriers to collaboration
 - Email “there’s a typo in your code in file X, line 30” vs
 - Pull request “here’s a correction to your code”
- Free for researchers and students

General idea

<https://jules32.github.io/2016-07-12-Oxford/git/>

REMOTE (aka Github website)



Clone (i.e., copy)
repository to your
computer (a one
time event)

Pull remote
changes

Push local
changes



LOCAL
(aka your computer)

I SHOULD USE GITHUB



More good practices

(i.e., version control is one of many)

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For matting and readability is important both when writing text and when writing code

Formatting and readability are important both when writing text and when writing code

Tidyverse style guide

AUTHOR

The tidyverse team

Welcome

Good coding style is like correct punctuation: you can manage without it, but it sure makes things easier to read. This site describes the style used throughout the [tidyverse](#). It was derived from Google's original R Style Guide - but Google's [current guide](#) is derived from the tidyverse style guide.

All style guides are fundamentally opinionated. Some decisions genuinely do make code easier to use (especially matching indenting to programming structure), but many decisions are arbitrary. The most important thing about a style guide is that it provides consistency, making code easier to write because you need to make fewer decisions.

<https://style.tidyverse.org/>



Key principle: Computational empathy

- *Keep in mind: The replication package is meant to be run by others, who have none of the setup, packages, and data that the original author might have, on computers that may not run the same operating system.*
- *Treat the replication package as one of the methods to convey the methods that lead to your manuscript's conclusions. Consider it a teaching tool, targeting young graduate students who may not be in your field.*

<https://aeadataeditor.github.io/aea-de-guidance/preparing-for-data-deposit.html>



Good ideas (programming language agnostic)

- Avoid doing things by hand or through point and click
- “I’m only doing this once...”

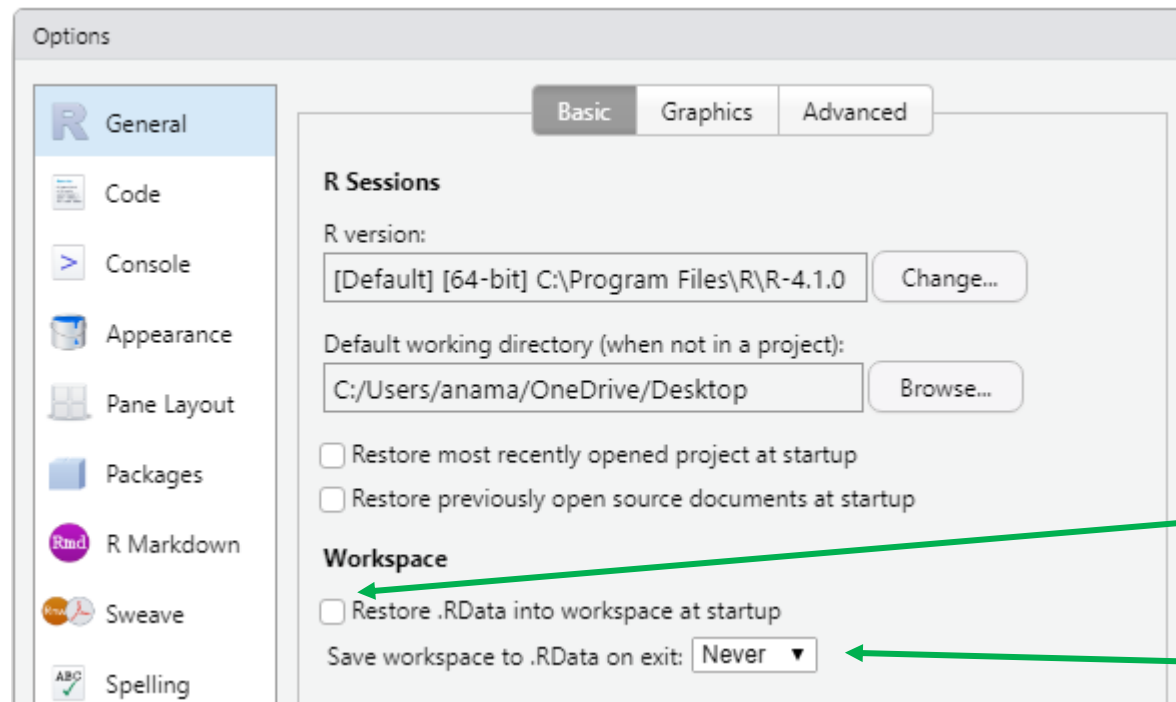
Things done by hand need to be precisely documented (harder than it sounds).

Good ideas (programming language agnostic)

- Save the data + code that generated the output, rather than only the output
- Make a deliberate choice to save (or not) intermediate output. Intermediate files can be ok as long as:
 - there is clear documentation of how they were created and
 - the links to and from these files are working

Good ideas, the R & RStudio edition

Rstudio -> Tools -> Global Options



Do NOT restore .RData at startup

Do NOT save .RData on exit

Good ideas, the R & RStudio edition

- Use R Projects and the ``here`` package
 - <https://tidyverse.org/blog/2017/12/workflow-vs-script/>
- Do NOT use ``setwd`` in your code
- Do NOT use ``rm(list = ls())`` in your code
- Format your code
 - Check Rstudio -> Code -> Reindent and Reformat

Good ideas, the R & RStudio edition

- Check section Recommendations (p. 4-5) in the Reproduction Report at <https://osf.io/mepbv/overview>
 - This was part of an i4R NHB collaboration, where we completed a reproducibility check of an NHB paper (Grossman et al. 2023, <https://doi.org/10.1038/s41562-022-01517-1>). The response from the authors of the NHB paper is included in the same OSF project
 - Code and data available at: https://github.com/Meta-Science-Replication-Team/repro-Forecasting_Tournament

Errors. Now what?



Learning resources

Where to find information about good practices

- Journal websites
- Published papers and books
- Free online learning resources
- Open Science communities, Reproducibility Networks

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Improving computational reproducibility in the social sciences

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Community Page

Best Practices for Scientific Computing

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PERSPECTIVE

Good enough practices in scientific computing

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How to Share Data for Collaboration

Shannon E. Ellis & Jeffrey T. Leek

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The American Statistician

ISSN: 0003-1305 (Print) 1537-2731 (Online) Journal homepage: <https://www.tandfonline.com/loi/utas20>

Excuse Me, Do You Have a Moment to Talk About Version Control?

Jennifer Bryan

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To link to this article: <https://doi.org/10.1080/00031305.2017.1399928>

20th ANNIVERSARY EDITION

The Pragmatic Programmer

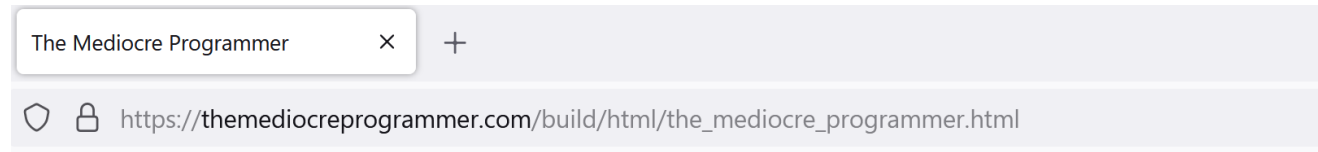

your journey to mastery

DAVID THOMAS
ANDREW HUNT

Foreword by SARON YITBAREK



https://themediocreprogrammer.com/build/html/the_mediocre_programmer.html



The Mediocre Programmer

Craig Maloney

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 - [0.2 Why this book?](#)
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 - [2.2 Backstage vs. performance](#)
 - [2.3 The lure of the post-mortem](#)

<https://osf.io/zmu3k/>

 OSFHOME ▼

 Workshop—GitFun: Introduction to git ... Files Wiki Analytics Registrations

Society for the Improvement of Psychological Science
(SIPS) 2021 Meeting /

Workshop—GitFun: Introduction to git and
GitHub

Contributors: [Ana Martinovici](#)

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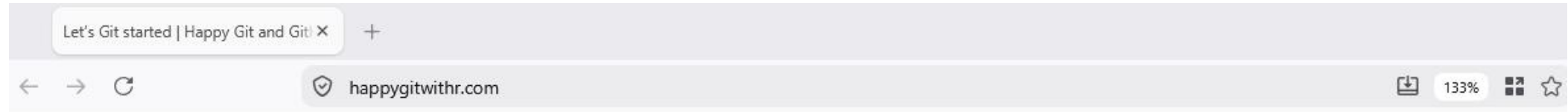
Identifier: DOI 10.17605/OSF.IO/ZMU3K

Category:  Software

Description: Slides and recording of the June 25, 2021 workshop.

License: Other 

<https://happygitwithr.com>



Happy Git and GitHub for the
user

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Let's Git started

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Installation

Let's Git started



Q&A

Thank you!